

Federico Mazzone

Cork, Ireland · mazzoni.federico1@gmail.com · [LinkedIn Profile](#) · [GitHub Portfolio](#) · [Scholar](#)

AI Engineer with experience building and deploying production LLM, RAG, and automation systems on AWS. Delivered customer-facing and internal AI products that increased engagement and reduced workflows from days to hours. Background in NLP and machine learning, with 8 peer-reviewed AI papers. EU citizen based in Ireland.

EXPERIENCE

AI Engineer

August 2026 – Present

Marriott International

Cork, Ireland

- **Multi-agent Assistant:** RAG-based Q&A, human escalation, and advance booking for Marriott's AI assistant.

AI Engineer

May 2024 – July 2026

Domotz

Dublin, Ireland

- **Chatbot:** Engineered a FastAPI RAG-based chatbot (using Qdrant, LangChain, PostgreSQL) deployed on AWS, boosting customer engagement by 65% and serving 100+ daily users. Built multiple knowledge bases and UIs.
- **Dynamic MCP:** Developed an MCP server acting as a semantic proxy across 10+ MCP servers, dynamically providing only the tools and skills relevant to each user's task via hybrid semantic search.
- **Conversation Intelligence:** Developed Gong alternative for Sales & CS teams, with classification, summarization, and chat over thousands of emails, tickets, and call transcripts, saving ~\$50k/year in subscription costs.
- **Automation Systems:** Built automation systems that reduced sales, finance, and marketing workflows from days to hours, including a Slack sales bot and invoice processors handling hundreds of customers.
- **Anomaly Detection:** Combined traditional ML and LLM reasoning to detect network anomalies in time series.

PhD Research Fellow in AI

Nov. 2022 – May 2024

University of Pisa / National Research Council of Italy

Pisa, Italy

- **Research Focus:** Collaborative human-AI systems, algorithmic fairness, linguistic trust in social networks.
- Published peer-reviewed articles, managed multiple concurrent projects from concept to validation, mentored university students, presented research at conferences, and organised the *Law, Society, and AI* workshops.

Freelance Data Analyst

2022

Self-Employed

Pisa, Italy

- Built recommendation systems, scraped web data for datasets, and explored bioinformatics biomarker methods.

PERSONAL PROJECTS

iRAG: RAG system that learns from human decisions, reducing error while gradually taking on more autonomy.

StockAgent: Multi-agent AutoGen system for stock market analysis with a custom MCP yfinance server.

HINTT: Real-time OCR+translation tool sending screenshots to OpenAI/Ollama. Employed tkinter and mss.

Simple Palette Maker: Streamlit app to colorize black-and-white GameBoy palettes. 1k+ users.

SKILLS

AI/ML Stack: LLMs (LangChain, LangGraph, AutoGen, Bedrock), RAG, HuggingFace, SK-Learn, PyTorch, RiverML

Backend & Deployment: FastAPI, Streamlit, Docker, AWS (EC2, S3, CloudWatch, Terraform), PostgreSQL, Qdrant

Programming: Python, JavaScript, R

Platforms: Git, Jira, BitBucket, Snowflake, Zendesk, Salesforce, n8n, OpenAI, Claude, Amazon Bedrock, Azure

Languages: Italian (Native), English (C1), German (A2)

EDUCATION

University of Pisa

2020 – 2022

Master's Degree in Digital Humanities (Data Science and NLP) — 110/110 with honours (GPA: 4/4)

Pisa, Italy

- Relevant Courses: Data Mining, Social Network Analysis, Text Analytics, Distributed Data Analysis and Mining, Semantic Web, Laboratory of Data Science, Statistics, Applied Linguistics, Computational Linguistics.

University of Pisa

2014 – 2017

Bachelor's Degree in Philosophy — 110/110 with honours (GPA: 4/4)

Pisa, Italy